Lecture : Bit Manipulation - 1

Agenda

- Basics of logical operators
- Power of left shift operators
- Check ith bit is set ?
- Total number of set bits in n.
- Unset ith bit of a number.
- Set bits in a range.

Truth table for bit-wise operators.

a	b	a & b	a | b	a ^ b	~ a
		0 dominates	1 dominate	same same puppy shame	
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

& $\longrightarrow$ And

| $\longrightarrow$ OR

^ $\longrightarrow$ xor

~ $\longrightarrow$ not

AND properties

1.> Even | odd number

In binary representation —

Even number $\Rightarrow$ Lsb $\Rightarrow 0$

odd number $\Rightarrow$ Lsb $\Rightarrow 1$

35	100011 & 1 = 1
63	111111 & 1 = 1
7	111 & 1 = 1
9	1001 & 1 = 1

8	1000 & 1 = 0
18	10010 & 1 = 0
24	11000 & 1 = 0
26	11010 & 1 = 0

n & 1 → 1 → odd

n & 1 → 0 → Even.

```
if ( A & 1 == 0) {
      even
} else {
      odd
}
```

```
if(n % 2 == 0)
      Even
else
      odd.
```

2.> $A \,\&\, 0 = 0$

3.> $A \,\&\, A = A$

$$
\begin{array}{rl}
A = & 1\,0\,0\,0\,1\,1 \\
\& & 1\,0\,0\,0\,1\,1 \\
\hline
A = & 1\,0\,0\,0\,1\,1
\end{array}
$$

OR properties

1.) $A \mid 0 = A$

$$\begin{array}{rl} A = & 1\,0\,0\,1\,0\,1 \\ \mid & 0\,0\,0\,0\,0\,0 \\ \hline & 1\,0\,0\,1\,0\,1 = A \end{array}$$

2.) $A \mid A = A$

$$\begin{array}{rl} A = & 1\,0\,0\,1\,0\,1 \\ \mid & 1\,0\,0\,1\,0\,1 \\ \hline & 1\,0\,0\,1\,0\,1 \end{array}$$

XOR properties

1.) $A \,^\wedge\, 0 = A$

$$\begin{array}{rl} A = & 1\,0\,0\,1 \\ \wedge & 0\,0\,0\,0 \\ \hline A = & 1\,0\,0\,1 \\ \hline \end{array}$$

2.) $A \,^\wedge\, A = 0$

$$\begin{array}{rl} A = & 1\,0\,0\,1 \\ \wedge & 1\,0\,0\,1 \\ \hline & 0\,0\,0\,0 \\ \hline \end{array}$$

Commutative property

$A \,\&\, B \Rightarrow B \,\&\, A$

$A \mid B \Rightarrow B \mid A$

$A \,^\wedge\, B \Rightarrow B \,^\wedge\, A$

Associative property

$(A \,\&\, B) \,\&\, C \Rightarrow A \,\&\, (B \,\&\, C)$

$(A \mid B) \mid C \Rightarrow A \mid (B \mid C)$

$(A \,^\wedge\, B) \,^\wedge\, C \Rightarrow A \,^\wedge\, (B \,^\wedge\, C)$

$a \wedge b \wedge a \wedge d \wedge b \Rightarrow \underbrace{a \wedge a}_{0} \wedge \underbrace{b \wedge b}_{0} \wedge d$

$\underbrace{0 \wedge 0}_{} \wedge d$

$0 \wedge d = \underline{d \text{ Ans}}$

$\cancel{1} \wedge \cancel{3} \wedge \cancel{5} \wedge \cancel{3} \wedge 2 \wedge \cancel{1} \wedge \cancel{5} \Rightarrow \underline{2 \text{ Ans}}$

Left shift operator (Multiplication)

$a << n \Rightarrow a * 2^{n}$.

$1 << n \Rightarrow 1 * 2^{n} = 2^{n}$.

$2 << 3 \Rightarrow 2 * 2^{3} = 2 * 8 = \underline{16 \text{ Ans}}$

Right shift operator (Division)

$a >> n \Rightarrow \frac{a}{2^{n}}$.

$1 >> n \Rightarrow \frac{1}{2^{n}}$

$16 >> 2 \Rightarrow \frac{16}{2^{2}} = \frac{16}{4} = \underline{4 \text{ Ans}}$

Quiz $1 << 3 \Rightarrow \underline{1 * 2^{3} = 8 \text{ Ans}}$

Power of left shift operator {Amazon}

Property 1 And

$n = 45$ 1 0 1 1 0 1

case1:

	5	4	3	2	1	0	
$n = 45 \longrightarrow$	1	0	1	1	0	1	
$1 << 2 \longrightarrow$ &	0	0	0	1	0	0	
	0	0	0	1	0	0	$\longrightarrow 4$

$2^2 = 4$

case2:

$n = 45 \longrightarrow$	1	0	1	1	0	1	
$1 << 3 \longrightarrow$ &	0	0	1	0	0	0	
	0	0	1	0	0	0	$\longrightarrow 8$

$2^3 = 8$

case3:

$n = 45 \longrightarrow$	1	0	1	1	0	1	
$1 << 4 \longrightarrow$ &	0	1	0	0	0	0	
	0	0	0	0	0	0	$\longrightarrow 0$

$2^4 = 16$

⋮

$n \;\&\; (1 << k)$

- non-zero — kth bit = 1
- 0 — kth bit = 0

TC: $O(1)$
SC: $O(1)$

Property 2

$n = 45 \longrightarrow 1\ 0\ 1\ 1\ 0\ 1$

$1 << 1 \longrightarrow\ |\ 0\ 0\ 0\ 0\ 1\ 0$

$2^1 = 2$

$1\ 0\ 1\ 1\ 1\ 1$

$n = 45 \longrightarrow 1\ 0\ 1\ 1\ 0\ 1$

$1 << 2 \longrightarrow\ |\ 0\ 0\ 0\ 1\ 0\ 0$

$2^2 = 4$

$1\ 0\ 1\ 1\ 0\ 1$

$n = 45 \longrightarrow 1\ 0\ 1\ 1\ 0\ 1$

$1 << 3 \longrightarrow\ |\ 0\ 0\ 1\ 0\ 0\ 0$

$2^3 = 8$

$1\ 0\ 1\ 1\ 0\ 1$

$n = 45 \longrightarrow 1\ 0\ 1\ 1\ 0\ 1$

$1 << 4 \longrightarrow\ |\ 0\ 1\ 0\ 0\ 0\ 0$

$1\ 1\ 1\ 1\ 0\ 1$

⋮

$n \mid (1 << k)$

kth bit = 0 → final ans: kth bit = 1

kbit = 1 → same

Qu set ith bit of a no n.

Ex: $n = 45$
$i = 4$

5	4	3	2	1	0
1	0	1	1	0	1
1	1	1	1	0	1

$\Longrightarrow$ 61 Ans

Property 3   Toggling   xor

$n = 45 \longrightarrow$ 1 0 1 1 0 1

$\boxed{1 << 1} \longrightarrow$ ^ 0 0 0 0 1 0

$2^1 = 2$

1 0 1 1 1 1

$n = 45 \longrightarrow$ 1 0 1 1 0 1

$1 << 3 \longrightarrow$ 0 0 1 0 0 0

2^3

1 0 0 1 0 1

$n = 45 \longrightarrow$

$1 << 4 \longrightarrow$

⋮

$n \hat{} (1 << k)$

toggling

Kth bit = 0   Kth bit = 1

Kth bit = 1   Kth bit = 0

$n = 45$

$i = 2.$

toggle ith bit

1 0 1 1 0 1

1 0 1 0 0 1 $\longrightarrow$ 41 Ans

Qu. Check whether ith bit is set (1) or not?

input $n = 45 \rightarrow$

5	4	3	2	1	0
1	0	1	1	0	1

$i = 2 \longrightarrow$ true

code

```
boolean checkSetBit(int n, int k) {
        if (n & (1<<i) == 0) {
                return false;
        }
        return true;
}
```

Qu. Count total numbers of set bit in n.

input

$n = 12 \longrightarrow 1100 = 2$

$n = 45 \longrightarrow 101101 \longrightarrow 4$

Approach 1

```
int countTotalsetBits (int n) {
    cnt=0;
    for(i=0; i<32; i++) {
        if ( checksetBit(n, i)) {
            cnt++;
        }
    }
    return cnt;
}
```

TC: $O(32) \simeq O(1)$

SC: $O(1)$

Approach 2

n = 45

$\begin{array}{cccccc} 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \end{array} \longrightarrow$ 3 Ans

n = 45

n = n>>1

n>>1

1 0 1 1 0 1 & 1 = 1

1 0 1 1 0 & 1 = 0

1 0 1 1 & 1 = 1

```
int countTotalsetBits(int n) {
    cnt = 0
    while (n > 0) {
        if (n & 1 == 1) {
            cnt++;
        }
        n = n>>1;
    }
    return cnt;
}
```

10^9

|

$\frac{10^9}{2}$

⋮

0

32

TC: O(1) $\log_2(n)$

SC: O(1)

Qu.3 Unset the ith bit of a number if ith bit is set. [toggle]

input: $n = 6$ 2 1 0
1 1 0

$i = 2$ 0 1 0 $\Rightarrow$ 2 Ans .

Approach

```
int unsetIthBit(int n, int k) {
        if ( checkSetBit (n, k)) {
                n = n^(1<<k);
        }
        return n;
}
```

Break: 10:30 – 10:40

Qu.4 A group of computer scientists is working on a project that involves encoding binary numbers. They need to create a binary number with a specific pattern for their project. The pattern requires A 0's followed by B 1's followed by C 0's. To simplify the process, they need a function that takes A, B and C as inputs and return the decimal value of resulting binary number. Can you help them by writing a function that can solve this problem efficiently. ?

no significance

Constraints

0 <= A , B , C <= 20

~~A = 20~~

$\boxed{\begin{array}{l} B = 20 \\ C = 20 \end{array}} \rightarrow$ 40 bit $\rightarrow$ long

input

A = 4

B = 3

C = 2.

pattern

requires A 0's followed by B 1's followed by C 0's.

8	7	6	5	4	3	2	1	0
0	0	0	0	1	1	1	0	0

$\Rightarrow$ 28 Ans

Approach

A = 4

B = 3

C = 2

long ans = 0

63	...	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0	...	0	0	0	0	0	0	0	0	0	0	0	0	0	0
											1	1	1		

{ c idx to B+C −1

```
long solve(int A, int B, int C) {
        ans = 0
        for(i = c; i<= B+C-1; i++) {
            ans = ans | (1<< i);
        }
        return ans;
}
```

TC: O(1)
SC: O(1)

Thankyou ☺

Doubts

A = 6
B = 8
C = 2

$(2^{B} - 1) << \frac{\log_2(B)}{C} = 0$ 1

256 − 1

$\boxed{255}$

0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 0 0 0 0

1 1 1 1 1 1 1 1

2^{B} Math.pow(2, b)

20

$O(\log_2(n)) = 1$

32 64

long = 64 << b b traversal O(1)

int
lor

Qu n → 2
m → $1 <= m <= 10^5$

$\boxed{n^m \% 1000}$ $\log_2(n)$

10^5

2

ns = 0

for (i = 0 ———— b−1)

ans = ans +

(1<<(b − i + c − 1

3 − 0 + 2

5

A = 4
B = 3
C = 2

2 3 4

3 − 0 + 2

3

3 − 1 + 2

4